# Expanding, Factorising & Solving Equations

Answer sheet · 26 marks total

## Section A — Expanding single brackets

A1. $\mathbf{3x + 15}$ (1)

A2. $\mathbf{12x - 6}$ (1)

A3. $\mathbf{-4x + 12}$ (1)

A4. $\mathbf{-2x - 7}$ (1)

## Section B — Expanding double brackets

B1. $x^2 + 5x + 3x + 15$ (1)   = $\mathbf{x^2 + 8x + 15}$ (1)

B2. $x^2 + 6x - 2x - 12$ (1)   = $\mathbf{x^2 + 4x - 12}$ (1)

B3. $x^2 - 3x - 4x + 12$ (1)   = $\mathbf{x^2 - 7x + 12}$ (1)

## Section C — Factorising

C1. $\mathbf{4(2x + 5)}$ (1)

C2. $\mathbf{3(5x - 3)}$ (1)

C3. Numbers multiply to 15, add to 8: 3 and 5 (1)   = $\mathbf{(x + 3)(x + 5)}$ (1)

C4. Numbers multiply to –24, add to –2: –6 and 4 (1)   = $\mathbf{(x - 6)(x + 4)}$ (1)

## Section D — Solving equations

D1. $4x = 16$ (1)   = $\mathbf{x = 4}$ (1)

D2. $3x - 6 = 15 \rightarrow 3x = 21$ (1)   = $\mathbf{x = 7}$ (1)

D3. $6x - 2x = 11 + 5$ (1)   $4x = 16$ (1)   = $\mathbf{x = 4}$ (1)

D4. $4x + 6 = x + 21$ (1)   $3x = 15$ (1)   = $\mathbf{x = 5}$ (1)

*Marking note: award method marks for correctly expanding brackets or correctly identifying the factor pair, even if the final answer contains an arithmetic error. Accept equivalent forms (e.g. (x+4)(x+5) or (x+5)(x+4)).*